

PlacedOn — The Report Model: Algorithm, ML/NN Architecture, and the Two Reports

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You asked for a “deep thinking model” that, after the AI interview, produces a report with company fit, selection %, AI recommendation, pros/cons, what to improve, skills to develop, and a roadmap (courses / exams / colleges) — plus the ML/NN + LLM that runs *during* the interview. This is my CTO thinking on how to build it **so it’s powerful and survives a compliance review**, not just impressive in a demo.

0. The reframe that makes all of this work

Your wishlist is actually **two reports for two different people**:

	Employer Evidence Card (built)	Candidate Growth Report (new)
Owner	shared <i>by the candidate’s consent</i>	candidate-owned, candidate-facing
Answers	”what did they <i>prove</i> ?”	”where do I fit, what’s my edge, what do I fix, what’s my path?”
Contains	verified, quoted evidence · no score	fit, strengths, gaps, roadmap, courses/exams
Decision	the human decides	guidance to the candidate

Almost everything new you listed — company fit, advantages/disadvantages, what to improve, skills to develop, the roadmap — **belongs to the Candidate Growth Report**, because it’s *advice to the candidate*, not a verdict on them. That single split is what lets us build the ambitious version safely.

1. Two red lines I will not cross — and the honest substitute for each

Red line 1 — “% chance you’ll be selected at Company X.” I will **not** build a selection-probability number, for three reasons:

- **It’s not honestly computable.** Real selection depends on the applicant pool, the recruiter’s mood, headcount, budget, timing — none of which we observe. Any number is fiction.
- **It’s legally radioactive.** A model predicting a hiring outcome is an Automated Employment Decision Tool (NYC LL144, EU AI Act high-risk).
- **A wrong number does real harm** — false hope or false discouragement to a real person.

Honest substitute → “**Evidence-Fit / Readiness %**”: how much of a role’s *public* requirements (O*NET/ESCO + the company’s Role DNA if public) your evidence covers, with the exact gaps named. It answers “*Am I ready? What’s missing?*” — never “*Will they pick me?*”

Red line 2 — “**the AI recommends and will select.**” The AI **never** selects. It recommends *to the candidate* which roles their evidence fits best (career guidance) and *informs* the employer with cited evidence. **The human always decides.** This is not timidity — it’s the entire trust proposition.

Everything else you asked for is not just allowed; it’s the strongest, most defensible, most *candidate-loving* part of the product.

2. The Candidate Growth Report — your wishlist, mapped

You asked for	We build (defensible form)
Company fit / which company to apply	Evidence-Fit % per role family + specific roles/companies whose Role DNA your evidence matches, ranked by coverage — each with its gaps. “Apply here; here’s what’s strong; here’s what to shore up first.”
”% selection”	Readiness % (coverage of public requirements), never selection odds.
AI recommendation	”Your evidence fits these role families best” — guidance, not a decision.
Advantages	Your Supported items that clear the role’s typical bar or are rare in the taxonomy.
Disadvantages	Reframed as growth gaps : required skills with weak/no evidence.
What to improve	For each gap: the specific next evidence to build + a suggested project/behavior.
Skills to develop	Taxonomy-adjacent skills (O*NET “related skills” graph) that unlock more roles.
Roadmap: courses / exams / colleges	Sequenced plan from a <i>curated, cited catalog</i> mapped to skills — courses, certifications/exams, and formal programs. Never hallucinated by the LLM.

Plus: it’s **reproducible, candidate-owned**, and the candidate can **re-interview to upgrade a band** — the growth loop that creates stickiness without any social feed.

3. The “deep thinking model” is a PIPELINE, not one monolith

A single giant neural net that “reads the interview and outputs a hireability score” is exactly the thing we must not build (§4). The real system is a chain of small, inspectable models — each doing one job, each auditable.

INTERVIEW-TIME (real-time, adaptive) — mostly built —

- Adaptive questioner (LLM generator + planner/question_strategy)
 - | asks a base question → scores which Role DNA signals still lack evidence
 - | → an INFORMATION-GAIN loop picks the next probe = "basic → deeper"
 - | ("deep connection" = a running evidence-coverage state per signal;
 - | the next question is the one that reduces the biggest uncertainty)
- ▼
- Turn compressor + trust triggers (built)

POST-INTERVIEW (report generation) — the new stack —

1. Extractor LLM, schema-constrained: transcript → {claim, quote span, skill}
2. Fabrication gate verify_quote (deterministic, BUILT): quote must be verbatim
3. Fidelity judge cross-provider LLM: does the quote entail the claim?
- drop + alert faithful < 0.8 → drop · harmful_inference →
4. Skill encoder ★ SBERT embeds each observation + each 0★NET/ESCO/WEF skill;
- cosine similarity maps observation → skill (pgvector)
- ★ this is the real neural network – a transparent ENCODER, not a decider
5. Band calibration deterministic thresholds tuned against candidate reviews (trust loop)
6. Coverage + gaps deterministic: evidenced skills vs role requirements → Fit % + gaps
7. Growth synthesis retrieval + templates over a CITED catalog → roadmap/courses/exams
8. Reproducibility pin model/prompt/taxonomy versions + signature

Every arrow is inspectable; every number traces to a quote or a taxonomy edge.

4. The neural network — and why NOT one big hiring net (holding your own decision #3)

The legitimate neural network is the skill encoder (step 4): **SBERT sentence-embeddings** (you already depend on `sentence-transformers`) + **pgvector** retrieval. It maps a candidate's demonstrated behavior to standard skills by *meaning*, not keywords — so “I checked the idempotency keys” maps to *Troubleshooting* even without the word. It's a neural net used as an **encoder** (transparent, every match has a similarity score you can show), never as a hidden decider.

Why we do NOT train a proprietary hire/score neural net in V1 (this was your call, and it's right):

- **No labels.** You can't train a hire-predictor without a history of hires + outcomes.
- **Small data + deep nets = overfitting** and the death of explainability — and explainability *is* the moat.
- **Legal.** A learned accept/reject model is a high-risk AEDT.

The responsible ML path (staged):

1. **Now:** foundation LLMs as *constrained components* (extractor, judge, question-gen) + deterministic gates.
2. **Now/next:** the **SBERT encoder + pgvector** — the real, transparent NN.
3. **Later, only after outcome data** accrues from the **Slice 8 outcome check-ins**: a **LightGBM re-ranker** (gradient-boosted trees — the *right* tool for small tabular data, not a deep net) that **re-orders candidates for a human**, behind an **adverse-impact ≥ 0.80 gate**, and **never rejects**. That is the “ML that supports the AI,” built when it's honest.

5. How the interview gets “deeper” (the active-learning loop, concretely)

1. Role DNA defines the target signals (Slice 1). Each starts at *evidence = none*.
2. The questioner asks a base, open question for the highest-priority signal.

3. The candidate answers in text (or voice→text). The extractor + judge score *how much evidence* that answer gave for which signals.
4. The loop picks the **next question to maximize information gain** — probe the signal with the weakest evidence, or go **deeper** on a promising-but-thin one (“you said you rolled back — what did you check first?”).
5. Stop when coverage is sufficient or the turn budget is hit (existing `should_end_interview`).

That’s the “deep connection”: the model is continuously deciding *what it still doesn’t know about this candidate’s fit* and asking exactly that. It’s real, it’s mostly built, and it needs the hosted LLM to run live.

6. What else to add (research-informed) + my recommendation

- **Confidence/uncertainty on every number**, not just a point value — honesty compounds trust.
 - **Re-interview to upgrade a band** — the growth/engagement loop (no social feed).
 - **Percentile framing only within a role family, coverage-based** — never a person-vs-person rank.
 - **Evidence freshness** — skills decay; WEF says **39% of skills change by 2030** — show recency.
 - **Adverse-impact dashboard** from day one; **model/version logging** so any report is reproducible.
 - **Do NOT**: scrape social, infer protected attributes, predict selection odds, let the AI reject.
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7. Phasing (extends the report-card plan)

Phase	Adds
1	Taxonomy service (O*NET/ESCO/WEF) + role requirements + Card A schema (from the report-card plan)
2	Card B rendering + SBERT encoder + pgvector skill mapping (the NN)
3	Comparison engine + Candidate Growth Report (fit, gaps, roadmap) + curated course/exam catalog
4	Audience views (enterprise/startup densities) + signed verification artifact
5	Validation study + LightGBM re-ranker (gated by adverse impact; informs, never decides)

Hard dependency: steps 1–3 and the live active-learning loop need the **hosted backend + model keys**. The structure is deterministic and buildable now; the model-driven content is the seam — which is why the deploy is still the critical path.

8. My one-paragraph recommendation

Build the **Candidate Growth Report** as a first-class, candidate-owned product — that’s where “fit, strengths, gaps, roadmap, courses” live, and it’s genuinely novel and lovable. Keep the **employer** side to verified evidence with **no score and no selection odds**. Use the **SBERT encoder + pgvector** as the neural network now; defer any learned ranker until the outcome data exists and can pass a fairness gate. Do the two things that are uninferable-and-unethical (selection %, AI-decides) as their **honest substitutes** (readiness %, guidance). That’s the version that’s both more impressive *and* fundable.